

18. Fatal Accidents Accident data were analyzed to determine the numbers of fatal accidents for automobiles of three sizes. The data for 346 accidents are as follows:

	Small	Medium	Large	SUM
Fatal	67	26	16	109
Not Fatal	128	63	46	237
SUM	195	89	62	346

Do the data indicate that the frequency of fatal accidents is dependent on the size of automobiles? Test using a 5% significance level.

H_0 : frequency of fatal accidents is independent on the size of automobiles

H_a : the two factors are dependent.

$$\chi^2_{STAT} = \frac{\sum (O_{ij} - E_{ij})^2}{E_{ij}} \sim \chi^2_{0.05(3-1)(2-1)}$$

E_{ij}	S	M	L
F	$\frac{109 \times 195}{346} = 61.49$	$\frac{109 \times 89}{346} = 28.04$	$\frac{109 \times 62}{346} = 19.53$
NF	$\frac{237 \times 195}{346} = 133.51$	$\frac{237 \times 89}{346} = 89$	$\frac{237 \times 62}{346} = 62$

$$\chi^2 = 1.884$$

$$RR = \{ \chi^2 \mid \chi^2 > 5.991 \} \quad 1.884 < 5.991 \Rightarrow \text{non Reject } H_0$$

$\chi^2_{1-0.05, 2} = 5.991$ There isn't sufficient evidence to say that two factors are dependent.